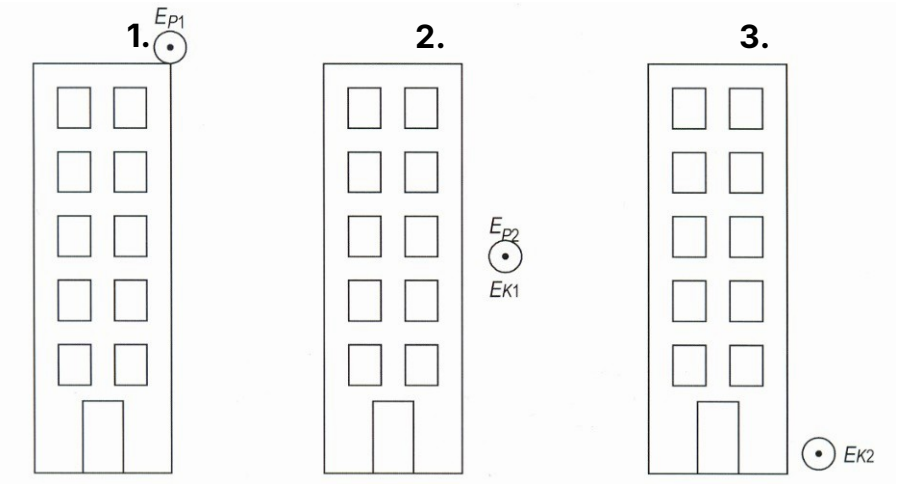
## Calculating energy transfers

It has already been stated that energy cannot be created or destroyed – it can only be transformed to another energy. During an energy transformation, the total energy contained within any closed system remains constant. Knowing the total amount of energy at the start (or end) of any energy transformation tells us the total energy at any given time during the transformation.

### Example

**1.** Consider a ball that is released from a high building. When the ball is at the top of the building, it has gravitational potential energy ($E_{P1}$)




**The total energy, $E_T$ = $E_{P1}$**

**2.** When the ball is released, it falls and some of the potential energy ($E_{P2}$) is converted into kinetic energy ($E_{K1}$). The total amount of energy ($E_T$) the ball possesses at this time is equal to the potential energy plus the kinetic energy.

**The total energy $E_T = (E_{P2} + E_{K1})$**

**3.** Just as the ball hits the ground, it no longer has any potential energy – all the potential energy ($E_{P1}$) has been converted into kinetic energy ($E_{K1}$). The total amount of energy now consists of the kinetic energy alone.

**Total energy $E_T = E_{P1} = (E_{P2} + E_{K1}) = E_{k2}$**

## Worked example: transfer of energy (potential to kinetic)

A body of mass 30kg falls freely from a height of 20 metres. Find its final velocity and kinetic energy at impact. First calculate the initial potential energy.

$E_P = mgh$

$= 30 \times 9.8 \times 20$

$= 5880J$ = **5.9 kJ** (to 2 significant figures)

This potential energy is converted or transferred into kinetic energy, which means that the kinetic energy at impact is equal to 5.9 kJ. But this is not the end of the calculation!

To calculate the final velocity of the body we begin by taking $E_k$ = 5.88 kJ.

$E_K = ½mv^2$

$5.88 \times 10^3 = ½ \times 30 \times v^2$

$5880 = 15 \times v^2$

$v^2 = 5880 / 15$

$v^2 = 392$

$v =$ = 19.79 m/s = **20m/s** (to 2 significant figures)

This type of calculation can be completed for any type of energy conversion. Knowing the total amount of energy at any given time (beginning/middle/end) is the same at any other given time.

**Q1.** At the top of a 50cm ramp a 1.5kg buggy rolls down to the bottom using gravity. Work out the potential and kinetic energy at the top and middle of the hill. Calculate the transfer of energy.

h = 0.5m (top) m = 1.5kg g = 9.8 N/kg or 9.8 $ms^{-2}$

(top) (middle)

$E_{p\ TOP}$ = 7.35J $E_{p\ MIDDLE}$ = 3.675J

$E_{K\ TOP}$ = 0 J $E_{K\ MIDDLE} = E_{p\ TOP} - E_{p\ MIDDLE} = 7.35 - 3.675 =$ 3.675J

Therefore, **3.7J** of energy has been transferred from potential to kinetic energy (to 2 sig. fig.)

**Q2.** At the top of a 80cm ramp, a 3kg buggy rolls down to the bottom using gravity. Work out the potential and kinetic energy at the top and middle of the hill.

h = 0.8m  m = 3kg  g = 9.8 N/kg or 9.8 $ms^{-2}$

$E_p$ = 23.5J at the top

$E_{p\ TOP}$ = 23.52J  $E_{p\ MIDDLE}$ = 11.76J

$E_{K\ TOP}$ = 0 J  $E_{K\ MIDDLE} = E_{p\ TOP} - E_{p\ MIDDLE}$ = 23.52 –11.76 = 11.76J

Therefore, **12J** of energy has been transferred from potential to kinetic energy (to 2 sig.fig)

**Q3.** A 5kg mass is raised steadily through a height of 2m. What work is done and what is the body's potential energy relative to the start?

$E_w = Fd$

F=ma (mass x accelereation) acceleration in this case is gravity, replace a with g.

F = mg
= 5 × 9.8
= 49N

then work done = F.d (force x distance)
= 49 × 2
= **98J**

Or calculate by:

$E_P = mgh$
$E_P = 5 \times 9.8 \times 2$
$E_P$ = **98J**

In other words, the work done (energy) required to lift the mass to that height is the same as the potential energy at that height.

**Q4.** A mass of 20kg is allowed to fall freely from a certain height above a datum. When the body is 16m above the datum, it possesses a total energy of 3531J. What is the starting height of the object?

Two methods can be used to solve...

$E_p = mgh$ $E_{p \text{ at 16 metres}} = mgh = 20 \times 9.8 \times 16 = 3{,}136J$

$E_{TOTAL \text{ at 16 metres}} = 3{,}531J$ (this is a combination of potential and kinetic energy)

Therefore, potential energy converted to $E_{k \text{ at 16 metres}} = 3531 - 3136 = 395J$

This kinetic energy, 395J, must have been converted from potential energy so to calculate the extra height above 16 metres;

$E_{p \text{ extra}} = mgh$

$E_{p \text{ extra}} / (mg) = h$

$h = 395 / (20 \times 9.8)$

$h = 2.015306$ m

$h_{\text{starting height}} = 16 + 2.015306 =$ **18 metres** to 2 sig. fig

Or more simply, we know that the total energy is 3,531J at 16 metres. This must be the combined potential energy and kinetic energy. Since there are no losses and potential energy converts to kinetic energy, we can simply assume that at the start of the fall 3,531J was the total potential and kinetic energy combined. However the mass is stationary at the start of the fall, so kinetic energy is zero. Therefore,

$E_p = mgh$

$E_p / (mg) = h$

$h = 3531 / (20 \times 9.8)$

$h_{\text{starting height}}$ = **18 metres** to 2 sig. fig. The same answer as before but a lot less working!!